VICTORIA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2024
H1 GEOGRAPHY PAPER 1 8834/01

MARK SCHEME

Section A

1		**Resource 1 shows the profile of the 56 participants. Resource 2 shows the survey recording sheet. Resource 3 shows the results of selected survey questions.**	
	(a)	**Suggest a possible hypothesis for the investigation and explain why the hypothesis is appropriate.**	**[3]**
		Award 1 mark for suggested hypothesis. Award 1 mark for each explanation of why the hypothesis is appropriate, to a maximum of 2 marks. Possible responses include: -A possible hypothesis: "Good physical health is the most important in order to age well among elderly living in Winterthur, Switzerland." [1] -The hypothesis is appropriate because there is a clearly defined/specific target group identified for the investigation which are the elderly [1] -The hypothesis is appropriate because it is at a suitable scale as the investigation is confined within Winterthur, Switzerland instead of the whole Switzerland [1] -The hypothesis is appropriate because it clearly defines the specific elderly need to age well which is "good physical health" [1] **AO1**	
	(b)	**Explain two criteria which the students could use to select participants for the investigation.**	**[4]**
		Award 1 mark for each explanation of a criterion the students could use to select the participants for their face-to-face interview. Award a maximum of 1 additional mark for further development of each criterion, where applicable. Possible responses include: -The participants must be elderly residents of Winterthur, Switzerland as this is the target group specified in the hypothesis [1 mark]. -The participants must be able to speak well without any speech impairments so that responses can be captured clearly without inaccuracies [1]. -The participants could be of differing employment status because being employed or not can influence an elderly's mental health OR being employed usually means being active mentally as the elderly go about with his/her tasks and responsibilities [1]. Furthermore, daily interactions or communicating or working with work colleagues can also aid in improving sense of well-being [1].	

		-The participants could come from different living arrangements as living alone or living with family or friends can have an influence on one's sense of well-being as the interactions with family or co-occupiers of the house can aid in achieving that [1]. **AO1**	
	(c)	**With reference to Resource 2, suggest how the survey recording sheet can be improved.** Award 1 mark for each suggestion on how the survey recording sheet can be improved. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible responses include: -The term 'diagnosed disease' is ambiguous hence the students can list the common elderly illnesses to guide the participants for the question [1]. These illnesses include diabetes, hypertension or musculoskeletal problems [1]. -The term 'use assisted aid' is ambiguous hence the students can list the common assisted aids which elderly require [1]. This includes reading glasses and walking aids such as walking sticks [1]. -The question on self-rated fitness may lead to the elderly inaccurately measuring their level of fitness without the guidance of a medical practitioner [1]. Students can provide detailed descriptors on what each level of fitness mean [1]. -The three themes are too broad which can cover many aspects [1]. They can put subcategories under each term so that the elderly can make a more informed choice e.g. sense of well being can cover aspects such as financial security, family networks [1]. -Participants are only allowed to choose one option for both questions hence, the students can allow participants which need they perceive to be important in ageing well [1] so that the students can gather data on the relative importance of the different options [1]. **AO2**	**[4]**
	(d)	**Explain the strengths of the students' approach to the interviews for the investigation.** Award 1 mark for each strength of the interviews. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible responses include: -Discussions were conducted in their Mother Tongue which is the language the elderly are comfortable with which can help the elderly to express their opinions with ease or without any misinterpretation [1] OR they can better	**[5]**

		understand each other and build on each other's opinions for a richer discussion [1]. -Discussions were conducted in familiar and comfortable surroundings including the participants' homes, hence, this relaxed environment enable them to share in a casual and relaxed setting which can allow for freedom of expression [1]. -Using a voice recorder allows for the points raised during the interview to be captured accurately [1]. **AO2**	
	(e)	**Explain how the students could minimize two potential ethical issues which might arise from the investigation.** Award 1 mark for each suggestion on how the students could minimize potential ethical issues from the investigation, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each suggestion, where applicable. Possible responses include: -Sharing confidential information, especially obtaining information on marital status, diagnosed disease or self-rated fitness level. This can be minimised by getting the elderly to acknowledge a confidentiality clause which states that the elderly are willing parties of the investigation and willing to share in front of other participants [1]. The clause may also include the "promise" that the information will only be used for research purpose and will not be sold to third parties [1]. -Recording the responses of the participants using a voice recorder is an issue as there is no anonymity and researchers can attribute/recognise who said what. To minimise this, researchers can inform the participants the recording procedures before the start of the FGDs and give them an opt out option if they are uncomfortable with the recording process [1]. -Conducting the interviews in the homes of the participants is an issue as the participants may be engaged in their daily home activities which may be interrupted to make way for the interview. To minimise this, participants can be given the choice to conduct the interview at the homes or recommend a suitable and conducive place [1]. **AO1**	**[4]**

	(f)	**The students concluded that the elderly in Winterthur, Switzerland perceived having good physical health as the most important to age well. Evaluate the validity of their conclusion.** Possible Approaches Candidates could refer to the resources on how accurate and reliable the collected data are for drawing conclusion about what elderly perceived as the most important to age. Candidates could look at the way data is collected via surveys and interviews could make accuracy and reliability of the data questionable. I The students' conclusion is valid to a <u>small extent.</u> Reasons why the conclusion is valid: -Based on the survey results. the theme "having physical health" received the highest percentage out of the themes (50%) Reasons why the conclusion lacks validity: -These responses were based on responses of 56 participants, which is not representative of all the elderly living in Winterthur, Switzerland [Accuracy] -The questions were framed based on the three themes – there could be other themes/aspects which are also important in aging well [Accuracy] -The participants were recruited using convenience sampling who may have the same lived experiences as each other (e.g. same household, family members), hence the result may not be representative of all the elderly living in Winterthur, Switzerland [Accuracy] -There is no equal representation of men and women among the 56 participants (53 vs 3) – hence, responses may be too skewed to the female experiences [Accuracy] -A few interviews were carried out in pairs, which may result in participants' responses being swayed or influenced by each other's responses or having one participant overpowering the other. There is no consistency to how the interviews were carried out [Accuracy/Reliability] *Levels marked using Generic Level Descriptors for H2 fieldwork evaluative questions* **AO3**	[10]

Levels	Marks	Generic Level Descriptors for H1 Fieldwork Evaluative Question
3	9–10	Evaluation is analytical and coherent. Response addresses the question and demonstrates good knowledge and understanding of fieldwork methods relevant to the given context.
2	5–8	Evaluation is mostly analytical and coherent. Weaker responses in this level will have evaluation that is broadly analytical and generally coherent. Response generally addresses the question and demonstrates adequate knowledge and understanding of fieldwork methods relevant to the given context.
1	1–4	Response is descriptive with limited or no analysis and evaluation. Response is fragmented and lacks clarity. Response lacks focus on the question and may be largely irrelevant to it. Response shows basic knowledge and understanding of fieldwork methods. Response has some, though limited, relevance to the given context.
0	0	No creditworthy response

Section B

2 Resource 6 shows the changes in the Earth Overshoot Day from 1971 to 2024. Resource 7 shows waste disposed by types of settlements in Mthatha, South Africa. Resource 8 shows the locations of different types of settlements in Mthatha. Mthatha is a fast-growing city in South Africa.

(a) Explain what is meant by ecological footprint. **[3]**

Award 1 mark for each explanation of the components and/or measurement of ecological footprint, up to 3 marks.

Possible responses include:

Definition

- Measure of productive land and aquatic ecosystem needed by humans to produce the food, energy and water resources that we need for consumption, and absorb the wastes that human produces [1]
- The higher the ecological footprint, the more resources are needed to produce the resources we need and absorb the waste we produce. [1 – for the relationship]

Measurement and usage

- Can be measured in hectares and at different scales [1] it is measured in total hectares at the global scale, regional and national scale, and in hectares per person at the individual scale [1]
- When the EF exceeds the biological capacity, there will be a ecological deficit. OR When the EF is less than the biological capacity, there will be an ecological surplus. [1]

AO1

(b) With reference to Resource 6, describe the change in the arrival of the Earth Overshoot Day between 1971 to 2024. **[5]**

		Award 1 mark for each description of the change in the Earth Overshoot Day, to a maximum of 5 marks. No marks are awarded when no change is seen in the description. Award a maximum of 1 additional mark for each description for further development of each description, including specific years and months. Note: Terms like "fell sharply" and "rise" are not accepted because it does not describe the overshoot day. No marks are given if these terms are consistently used. Instead, students should use "came much earlier" and "came later" to describe the overshoot day. Possible points include: • [General] Between 1971 to 2024, Earth Overshoot Day has been arriving earlier. [1] • [Specific] In 1971, the Earth Overshoot Day took play in late December but in 2024, the Earth Overshoot Day took place in August. [1] • [General] The Earth Overshoot Day generally became earlier between 1971 to 2009, but it started to arrive at the same time each year from 2010 to 2024. [1] • [Specific] Between 1973 to 2004, the Earth Overshoot Day came earlier by 1 month every decade. It took place in late December in 1971-1972, in November between 1973-1986, in October between 1987-1998, in September between 1999-2004, and in late August between 2004-2009. However, for the past 14 years since 2010, the Earth Overshoot Day has been arriving in early August. [1] • An anomaly is seen when the Earth Overshoot Day came later compared to the previous year when it was expected to come at a later date. This is evidence in 1980-1983, 1989-1992, 2002, 2009 and 2020. [1] **AO2**	
	(c)	**With reference to Resource 7, compare the waste disposed by informal slum settlements and low-density residential settlements.**	**[4]**
		Award 1 mark for each comparison between waste disposed between upgraded informal residential settlements and low-density residential settlements, up to a maximum of 2 marks for either similarity or difference. Award a maximum of 1 additional mark for further development of each comparison, where applicable. Note: Students should use comparative terms – higher, most, largest, 2 times more etc. Possible responses include: Similarity • Food waste takes up the highest proportion of all waste in both type of settlements, and it is significantly higher than the second highest contributor to waste. [1] In upgraded informal residential	

		settlements, food waste constitutes 46.9% of all waste and this is about 2.5 times the disposable nappies which is the second highest contributor to waste. In informal slums settlements, food waste constitutes 70.5% of all waste and this is about 7 times the disposable nappies which is the second highest contributor to waste. [1] • Disposable nappies second highest contributor to the total waste in both type of settlements. [1] It makes up 18.8% in upgraded informal residential settlement and 11.5% in low-density residential settlements. [1] • Of all the waste present in both settlement types, plastic waste was the lowest contributor to the total waste. [1] It makes up 6.3% of all waste in informal slums settlements, and 1.6% of all waste in low-density residential settlements. [1] Differences • Glass did not contribute to any waste in the informal slums settlements, but it was the 3rd highest contributor to waste in low-density residential settlements, constituting about 8.2%. [1] • The contribution of waste in informal slums settlements is more evenly spread out across the different waste types but it is more asymmetrical and skewed towards food waste in low-density residential settlements. [1] • The contribution of waste in informal slums settlements is closer to the average waste profile for all household compared to the low-density residential settlements. [1] In informal residential settlements, the largest deviation was 8.3% for food waste and smallest deviation was 3.2% for glass. However in the largest low-density residential settlements, largest deviation was 15.3% for food waste and smallest deviation was 0.6% for disposable nappies. [1] **AO2**	
	(d)	**Explain the reasons for the development of slums in urban areas of countries at low levels of development.**	**[4]**
		Award 1 mark for each reason, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each reason, where applicable. Possible responses include: • **Rapid urban growth** due to rural-urban migration and high natural increase. This will increase the demand for housing in the city. [1] • **Lack of affordable housing** because the city authorities are unable to build affordable housing fast enough to cope with the burgeoning urban population. [1] • **Lack of government and private-sector investment** into low-income housing. Investment may be low because of it may not be the most profitable land use since the housing will not be sold at a high price. Instead, money is diverted to more profitable landuse such as commercial or retail. [1]	

		• **Lack of job creation in the formal sector results in the persistence of low wages in the informal sector.** Most of the urban labour force in countries at low levels of development, about 60% of the workforce, are employed in the informal economy. It is not regulated by the government, has unstable pay and irregular hours, and the absence of work welfare benefits. [1] This reduces urban dweller's ability to break free from urban deprivation and they have to continue to stay in slums as that is the only housing option that the can afford with their meagre pay. [1] • **Low income of new migrants**: New rural migrants who enter the city in search of job opportunities are unable to afford most formal housing because they have not saved up enough money and earning money would be their top priority. When people are unable to find affordable formal housing, they will choose to live in existing slums due to their low cost. [1 additional mark] • **Lack of tenure in slums deters slum dwellers from improving their homes and slum surroundings, causing living conditions to deteriorate**. Slums are illegal dwellings without a secure tenure, which means that slum dwellers do not have rights to stay on that land. With the fear of being evicted anytime, they do not have any incentive to improve their surroundings and would live with land and waste pollution. [1] • **Institutional incapabilities** due to the red tapes/excessive bureaucracy and long construction approval process for housing. This delays the completion of affordable housing, and limits the supply of them. [1] • **Lack of land in urban areas to build affordable housing**. Due to the limited land in the city area, there is high competition of land from various land uses such as commercial, residential, industries. [1] The government may prioritise using land for more profitable land uses such as private housing, business parks and shopping malls to maximise the value of the land. As a result, the building affordable housing is made less important, causing the supply of the affordable housing to remain low. [1 additional mark] **AO1**	
	(e)	**With reference to Resources 7 and 8, explain how Mthatha's challenges in managing non-hazardous solid waste can limit its progress towards sustainable urban development.**	**[6]**
		Award 1 mark for each challenge, up to a maximum of 2 marks for each challenge. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible responses include: **Political challenges** • Government may lack the manpower to enforce laws against illegal dumpsites. R8 shows that there are 18 illegal dumpsites in Mthatha, and when laws are not enforced, it encourages people to use them as a main means of waste disposal. → land pollution [1]	

- There is no single strategy that can work universally through Mthatha since the waste profile and socio-economic profile differs in the city. R7 shows that there are different waste profiles in different types of settlements where there are different socio-economic classes. [1] This suggests that the government may find it hard to decide what type of strategies to implement at a city-scale. [1]

Social challenges

- There may be a lack of public participation for the 3Rs.
  - Sorting of waste may not be engrained as a habit amongst the city residents so there may be a lack of social participation for recycling because there is no evidence of recycling bins R8. [1]
  - Furthermore, there could be a habit of using disposable nappies and people are less willing to use washable ones as disposable nappies are more convenient. This is evident in R7 where disposable nappies are the second highest contributor to waste in informal slums settlements, low-density residential settlements and the average household. [1]
  - Furthermore, higher income residents have the disposal income to purchase what they need and less willing to buy only what they require since they can afford to buy more and throw when there is excess. [1] This is evident in R7 where food waste is the largest contributor to total waste in low-density residential settlement where the highest income residents live in. This may come from excess food that is bought but not eaten before the expiry date, or buffet restaurants that prepare more than what is eaten. [1]
- Strong NIMBY syndrome (Not-in-my-backyard) where locals want to have more landfills but they oppose to having them near their place of residences due to the air and land pollution. R8 shows that one-third, 6 out of 18 illegal dumpsites are located in the upgraded high-density informal settlements where there is likely to be higher proportion of lower income population. [1] In contrast, only have 2 out of 18 illegal dump sites are located in low density formal settlement where higher residents live. [1]
- NIMBY: The introduction of incineration plants may be not well received by the residents due to the release of particulate matter, heavy metals, toxic gases and ash. Residents do not want to be exposed to these gases in on a daily basis. [1] R8 shows that Mthatha is a relatively small city of a 10km by 15km, so any form of emissions can reach most parts of the city easily. [1]
- Some of the residents living in the illegal dumpsite within the upgraded high-density informal settlements may rely on scavenging as a source of livelihood and forceful eviction can cause them to lose their homes and income. [1]

Economic Challenge

- Although R7 shows us that South Africa has the highest GDP amongst African countries, but the economic development is not evenly spread across the country. Within fast-growing cities like Mthatha, there may still be limited funding because the government may prioritise economic development over environmental development because they infrastructural and industrial expansion

		is more about to attract long term investment compared to waste management infrastructure that may short-term improvement. The government may not be willing to trade-off economic development. [1] **Environmental Challenge** • The land and water pollution is too widespread for the government to contain. R8 shows that there are only 2 waste management strategies in the city which is illegal dumping and large, open-top containers. All illegal dump sites are on land, and 3 are near the rivers, one in the far west of the city and 2 in North crest. This may mean that the leachate from dump sites have contaminated the soil and rivers and the scale of the problem is too large for the current government to contain, so they may put it off for the time being. [1] Secondly, the large, open-top containers may also attract pests like rats and cockroaches since the smell of the rubbish is not contained. [1] **AO2**	

3		Resource 9 shows carbon emissions inequality within between and within selected countries. Resource 10 shows changes to the global carbon cycle from 2011 to 2020. Resource 11 shows a type of response to climate change.	
	(a)	**Describe the percentage of carbon dioxide emissions by income groups as seen in Resource 7.**	**[4]**
		Award 1 mark for a description of a carbon emission inequality Award a maximum of 1 additional mark for further development of each description. Note: As inequality = difference, hence similarities are not accepted. Possible responses include: • There is an uneven carbon dioxide emission by income. / There is inequality in carbon dioxide emission based on income. [1] • The richer the population, the higher their percentage of global carbon dioxide emissions from individual consumption. [1] ○ The richest 10% of the world's population contribute 49% of global carbon dioxide emission, while the poorest 50% world's population contribute 10% of global carbon dioxide emission. [1] ○ The richest 10% of the world's population contribute 49% of global carbon dioxide emission, which is 49 times more than the poorest 10% of the world's population who contribute a mere 1% of global carbon dioxide emission. [1] ○ The middle to high-middle income population who are the richest 20%-50% of the population contributes 41% global carbon dioxide emission from individual consumption. [1] • The richest 20% of the population accounts for most of the global carbon emissions from individuals, at 68%. [1]	

		○ Despite so, the richest 11%-20% of the population contributed only 19% of the global carbon emissions, which is less than half of the of the richest 10% of the population who contributed 49% of the global carbon emissions. • There is little disparity in the between the poorest 50% of the population as there is a slight increment of 0.5% in contribution to global carbon dioxide emission from individual consumption between each decile. [1] However, between the middle band of the top 30% to 50% of population, there is a slightly larger increment of 3 to 4% in contribution to global carbon dioxide emission from individual consumption between each decile. [1]	
	(b)	**With reference to Resource 8, explain how humans have influenced the global carbon cycle from 2011 to 2020.**	**[4]**
		Award 1 mark for the overall imbalance to the global carbon cycle, to a maximum of 1 mark. Award 1 mark for each explanation on activities that increase carbon emission, to a maximum of 3 marks. Award 1 mark for each explanation on activities that decrease carbon sinks, to a maximum of 2 marks. Since the data is about CO_2, methane (CH_4) and other carbon compounds are not accepted for this answer. Possible answer include: **Overall influence:** Human activities have led to an imbalance in the global carbon cycle, because there is more carbon dioxide added into the atmosphere than sequestered and stored by vegetation/soil and ocean carbon stores, resulting in a surplus of 19 Gt CO_2/year in the atmosphere. [1] **Activities that increase carbon emissions (syllabus terms)** • Burn fossil fuels: Resource 10 shows that burning of fossil fuels contributes 35 Gt CO_2/year. Fossil fuels contain carbon from plants and animals that lived millions of years ago. [1] When the organisms died, slow geologic processes trapped their carbon and transformed it into these natural resources. Burning fossil fuels or power plants is another way this carbon can be released into the atmospheric reservoir quickly. [1] Fossil fuels are widely used in DCs and LDCs to produce electricity in urban areas and industries, and power transportations networks. [1] • Land use changes: Resource 10 shows that landuse change contributes 5 Gt CO_2/year. Landuse change refer to conversion of a forested land to land that is used form agriculture, industries and cities. When land is cleared and excavated, the carbon in the soil is exposed to the more oxygen. Through carbon oxidation, more CO2 is produced. [1] In addition, cement is a main construction material of buildings in industries and cities produces large amount of CO2. [1] **Activities that reduce carbon sinks**	

		• Deforestation: Resource 10 shows that vegetation uptakes/absorbs 11 Gt CO_2/year. Vegetation are carbon sinks because carbon dioxide is absorbed during the process of photosynthesis. [1] However, the long-term, large-scale clearance of forests reduce the number of trees. This reduces the ability of the area to sequester and store carbon efficiently, and leaves more anthropogenic CO2 in the atmosphere [1] **AO2**	
	(c)	**Explain the impact of changes to the global carbon cycle seen in Resource 8 on global temperatures and aquatic ecosystems.**	**[4]**
		Award 1 mark for the impact of global carbon cycle on temperature, to a maximum of 2 mark. Award 1 mark for the impact of global carbon cycle on aquatic ecosystem, to a maximum of 2 mark. Award a maximum of 1 additional mark for further development of each description. Note: Aquatic ecosystems refer to freshwater and marine ecosystems. But to focus will be on ocean marine ecosystem here because that is shown in Resource 10. Possible responses include: **Impact on temperature** • There will be higher concentration of carbon dioxide in the atmosphere, as seen from the net surplus of carbon dioxide added into the carbon store. About 50% of human's carbon dioxide contribution, 19 Gt CO2/year, has been absorbed by the atmosphere store. [1] • Carbon dioxide is a greenhouse gas and it will trap more/greater retention (school notes) of outgoing longwave radiation from the earth, preventing them from escaping to space. This will lead to a higher global temperature through the enhanced greenhouse effect. [1] **Impact on aquatic ecosystems** • Resource 10 shows how oceans absorbed 10 Gt CO_2/year, which is one-quarter of the global carbon dioxide emission. When carbon dioxide dissolves into the ocean, it can reduce the pH of water. This can lead to ocean acidification and marine life are not able to survive in ocean water with lower pH. [1] • Furthermore, the increased sea surface temperature of can stress corals and cause them to expel the algae, leading to mass coral bleaching especially in the tropics. [1] As corals are shelter, spawning grounds and protection for marine ecosystems like starfish and shrimps, the collapse of coral reef ecosystem can lead to decrease in marine biodiversity, at worse, extinction of endangered species. [1] **AO1**	

	(d)	**Explain how changes in ice sheets can affect global temperatures through positive feedback mechanisms.**	**[4]**
		Award 1 mark for the explanation of positive feedback mechanism, to a maximum of 1 mark. Award a maximum of 1 additional mark for further development of each description. Possible responses include: • Through the positive feedback mechanism, melting of ice sheets can cause further warming of the global temperatures, which results in more melting. • Due to global warming, the Arctic region in the Noth Pole and Antarctica in the South Pole has warmed significantly since the 1980s, more than 3 times the average. This has led to the rapid melting of ice sheets in the Greenland and Antarctic Ice sheets. [1] • As a result, there is less white surfaces from the ice, and there will be less albedo to reflect sunlight back to space. [1] • In contrast, there will be more dark surfaces from the greater expense of dark ocean waters and exposed soils, which will result in more absorption of sunlight onto Earth's surface. As a result, there is higher temperature. [1] • This will then result in more melting of ice sheets and higher temperature absorbed. [1] **AO1**	
	(e)	**Using Resource 9, explain the role of key actors in responding to climate change.**	**[6]**
		Award 1 mark for the explanation of firms, state and individual, to a maximum of 4 marks per role. Award a maximum of 1 additional mark for further development of each description. Role of firms: • Firms need to prioritise environmental development for long-term economic growth even though it may mean higher costs in the short run. [1] Cooperate with the national laws and pay the carbon tax or switch to towards low carbon options that are energy efficient instead of pulling out from the country and going to another location. [1] • Firms that work with clean technology, such as solar energy, carbon capture and storage technology, can continue to develop their low-carbon technologies so that more companies and countries can adopt them. [1] Role of state: • Spearhead the "Singapore Climate Action Plan" to encourage different levels of society – firms, individuals, government sectors - to play a part in reducing carbon emissions. [1]	

		• Mandate a carbon tax on businesses since 2019. By using the "polluter pays" principle, it forces companies to find ways to reduce their carbon emission, hence contributing to lower overall national carbon emissions. [1] • Strong political will to be the first Southeast Asian country to enforce it despite knowing that might deter investor to even lower cost areas in the region. [1] Furthermore, despite knowing that the gradual increase in carbon tax to S$50 to S$80 per tonne of emissions by 2030 might deter investors, the government maintains a strong political commitment to reducing carbon emissions. [1] • Attract firms to develop and deploy low carbon technology such as more efficient solar panels, so as to reduce overall national carbon emissions. [1] • Money collected from the carbon tax will go towards funding more companies to transit to green technology. This will encourage more companies to reduce their carbon emissions. [1] Individuals [not on syllabus but can infer from R10] • Carbon tax levied on energy companies will be transferred to consumer. As electricity prices are expected to become more expensive, individuals are encouraged to use less electricity and save energy, in order to save costs. [1] **AO2**	

Section C: Essay

Levels	Marks	Generic Level Descriptors for H1 Essays
4	11–13	Evaluation is analytical and coherent. Response is mostly well-supported by relevant material, including the appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
3	8–10	Evaluation is broadly analytical and generally coherent. Response is moderately well-supported by relevant material, including some appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
2	5–7	Response is largely descriptive with limited analysis and evaluation. Response is partly coherent and may lack clarity in parts. Response is poorly supported by relevant materials, including the limited use of examples. Response features inaccurate geographical knowledge and poor understanding of the subject content relevant to the question.
1	1–4	Response is descriptive with no analysis or evaluation. Response is fragmented and lacks clarity. Response consists of unsupported assertions. Response features largely inaccurate geographical knowledge and a lack of understanding of the subject content relevant to the question.
0	0	No creditworthy response

4	**'The most pressing issues faced by elderly living in cities relate to their psychological well-being.' Evaluate this statement. [13]**
	Candidates could approach the question by make a judgement on whether **psychological well-being** (emotional needs and loneliness) is more pressing than **economic well-being** (employability and ageism at the workplace, rising cost of living and medical costs) and **social well-being** (access to medical care, crime, mobility) of elderly living in the city through a consideration of two or more case studies. Candidates can consider how pressing the issue is by considering the scale of the issue and the solvability of it. They could also consider the level of development in the city, cultural practices (e.g. filial piety) that could enhance or diminish their sense of well-being, the elderly's health status, age group and family structure, which could affect their expectations of well-being. Levels marked using Generic Level Descriptors for 13m H1 essays. **AO3**
5	**'The liveability of cities is mainly influenced by environmental factors. Evaluate this statement. [13]**
	Candidates could approach this question by making a judgement on whether **environmental factors** (air quality, water quality, green spaces, climate) is more important than **economic factors** (cost of living, job opportunities, strong currency) and **social factors** (inequalities faced by gender, age, nationality, income and subsequent segregation, crime rates and protests, access to healthcare, education, culture) in influencing liveability of a place through a consideration of two or more case studies. Candidates can consider how the assessment of liveability is subjective and is highly dependent on the assessor. With different groups of people living in the city (e.g. elderly, women, migrants, locals), their varied needs and priorities would influence their expectations of liveability. Levels marked using Generic Level Descriptors for 13m H1 essays. **AO3**
6	**'The impacts of contemporary climate change on less developed countries are greater than those on developed countries.' Evaluate this statement. [13]**
	Candidates could approach this question by making a judgement on impacts of contemporary climate change affect less developed countries more than developed countries through a consideration of criteria such as a temporal scale of the impacts, geographic location, demographics, the ability of the country to recover from the negative impacts, ability of the country to maximise the positive benefits. Impacts that students can consider can be both negative and positive, such as changing crop yield and impact on food security, water scarcity, flooding (coastal, inland, fluvial - caused by sea level rise, increased rainfall, storm surge from tropical cyclones), spread of vector-borne diseases. Levels marked using Generic Level Descriptors for 13m H1 essays. **AO3**

	AO1 (25m)	AO2 (39m)	AO3 (36m)
1(a)	3		
1(b)	4		
1(c)		4	
1(d)		5	
1(e)	4		
1(f)			10
2(a)	3		
2(b)		5	
2(c)		4	
2(d)	4		
2(e)		6	
3(a)		4	
3(b)		4	
3(c)	4		
3(d)	4		
3(e)		6	
4,5,6			26
Total	26	38	30